

SOP-SA-02 — Backend Implementation

Kode	SOP-SA-02
Pemilik	Saitama (Senior Backend Engineer)
Revisi	1.0 · Berlaku 2026-05-29 · Reviu per kuartal
Referensi	SWEBOK (Construction), 12-Factor App, ISO/IEC 25010, tools/be-impl-checklist.md
COBIT	BAI03 (Managed Solutions Build)

1. Tujuan

Mengubah kontrak yang sudah di-lock (SOP-SA-01) jadi implementasi backend yang **lestari (sustainable)** — clean separation, test cukup, evidence-first — bukan first-aid coding "yang penting jalan".

2. Ruang Lingkup

- **Berlaku:** implementasi endpoint, business logic, integrasi, refactor BE.
- **Tidak berlaku:** desain kontrak (SOP-SA-01), desain skema (SOP-SA-04 + @shikamaru), prototype Fauxbase-only di FE.

3. Definisi & Istilah

- **Service layer** — lapisan business logic, dipisah dari handler (HTTP) & repository (DB).
- **Transaction boundary** — batas operasi DB yang harus all-or-nothing.
- **Sustain vs first-aid** — refactor proper vs tambal band-aid (Tata mandate).

4. Referensi

SWEBOK Software Construction, 12-Factor (config di env, log as stream, stateless), Tata mandate (sustain, no tambal-sulam, evidence-first, stack-lock Fauxbase).

5. Tanggung Jawab (RACI)

Aktivitas	R	A	C	I
Implementasi	Saitama	Saitama	@shikamaru (query), @kakashi (pattern)	@killua
Code review	@kakashi	@kakashi	—	Saitama
Integration test	@I	@I	Saitama	@kakashi

6. Prosedur

6.1 Pra-impl (gate masuk)

- 6.1.1 Konfirmasi **kontrak udah lock** (SOP-SA-01) + sign-off Killua. Kalau belum → **stop, balik ke SOP-SA-01**.
- 6.1.2 Konfirmasi **skema fit** (joint @shikamaru). Kalau butuh skema baru → SOP-SA-04 dulu.

6.2 Implementasi

- 6.2.1 **Clean separation**: handler (parse/validasi HTTP) → service (business logic) → repository (DB). Gak campur business logic di handler.
- 6.2.2 **Input validation di boundary** — schema-based (Zod/joi/ajv). Reject di pintu masuk, jangan biarin masuk dalam.
- 6.2.3 **Query parameterized** — never string-concat SQL (anti-injection, kontrol SA-C2).
- 6.2.4 **Transaction boundary** — operasi multi-statement dibungkus transaksi (no partial commit).
- 6.2.5 **Data SACRED** — mutation lewat SOP-SA-04 (soft delete, audit trail). **Decision point**: ada DELETE/overwrite? → wajib SOP-SA-04.
- 6.2.6 **Auth enforcement** — endpoint sensitif lewat SOP-SA-03.
- 6.2.7 **Logging + error handling** — lewat SOP-SA-05 (structured log req-id+user-id, error code table).
- 6.2.8 **Stack-lock**: prototype → Fauxbase, **bukan Express manual / mock manual** (kontrol KK-C4 Kakashi).

6.3 Test & self-QC

- 6.3.1 **Unit test** service logic + **integration test** endpoint (happy + edge + error path).
- 6.3.2 Jalankan be-impl-checklist — **exit criteria**: semua item .
- 6.3.3 **Evidence** — curl/Postman output + test pass (Tata mandate evidence-first).

6.4 Handoff

- 6.4.1 **Buat engineering note high-level** (kontrol SA-C7) — kalau unit kerja non-trivial, tulis eng-note ikut `../../kakashi/tools/eng-note-template.md`, simpan di `saitama/notes/YYYY-MM-DD-<topik>.md` (konteks + keputusan teknis + dampak ke @killua/@shikamaru). Trivial (one-liner) cukup di `log.md`. **Exit criteria**: eng-note ter-arsip + ditautkan di `log.md`.
- 6.4.2 Handoff @I (integration test) + kabarin @killua (consume).
- 6.4.3 **Decision point**: lolos self-QC + ada evidence + eng-note ter-arsip (kalau non-trivial)? → handoff ke review @kakashi (SOP-KK-01) → Pre-Tata Gate. Belum → fix dulu, **jangan lapor "beres" tanpa bukti**.

7. Pengecualian

- **Hotfix S1 (prod down)**: boleh impl cepat untuk contain, test + checklist nyusul < 24 jam (post-fix), wajib dicatat + RCA (SOP-KK-05).

8. Rekaman & Retensi

Rekaman	Lokasi	Retensi
Narasi impl + evidence (curl/test)	log.md	Permanen
Engineering note high-level (unit kerja non-trivial)	notes/YYYY-MM-DD-<topik>.md	Permanen
Timesheet	timesheet.md	Permanen
Verdict review	PR / log.md (Kakashi)	Permanen

9. Indikator Kinerja (KPI)

KPI	Rumus	Target
Escaped defect BE	# bug lolos ke Tata/prod ÷ total endpoint	≈ 0
Test coverage path	# endpoint ber-test happy+edge+error ÷ total	↑ → 100%
First-pass review rate	# PR approve sekali ÷ total PR BE	↑ tiap periode

10. Riwayat Revisi

Rev	Tanggal	Perubahan
1.0	2026-05-29	Terbit pertama